Hyunjoon Lee

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Education

- **Integrated M.S./Ph.D. in Artificial Intelligence** — *Mar. 2024 – Present*
  Seoul National University — Seoul, Korea
  ◦ *Advisor: Jaesik Park*
- **B.E. in Mechanical Engineering** — *Mar. 2018 – Feb. 2024*
  Sungkyunkwan University — Suwon, Korea
  ◦ GPA: 4.39/4.5
  ◦ *Summa Cum Laude* ***(1/131)***

Research Interests

• Vision-Language-Action Model / World Action Model • Harness • 3D Vision

Research in Progress

- **Vision-Free In-Hand Dexterous Manipulation via Force Sensing**
  Co-Author
- **Open-X-Tactile: Advancing Generalist Tactile Policy through a Unified Open Data Ecosystem**
  Contributor

Publications

* *denotes equal contribution*

International

- **Occluded 3D Object Reconstruction via Masked Multi-view Volumetric Transformer**
  Jinmo Kim, **Hyunjoon Lee**, Namtae Kim, Eunsung Cha, Seungha Kim, Jaesik Park
  In Submission
- **Beyond the Current Scene: Event-Referential Grasping with Active View Selection**
  Hyunjoon Lee*, Haebeom Jung*, Eunsung Cha, Daeun Lee, Yu-Chiang Frank Wang, Jaesung Choe, Jaesik Park
  *In Submission · Memory for Robot Foundation Models Workshop @ Conf. on Robot Learning (**CoRL**), 2026* | link
- **Relation-Centric Open-Vocabulary 3D Gaussian Segmentation**
  Eunsung Cha*, **Hyunjoon Lee***, Jaesik Park
  *Winter Conf. on Applications of Computer Vision (**WACV**), 2027* | link
- **CF3: Compact and Fast 3D Feature Fields**
  Hyunjoon Lee, Joonkyu Min, Jaesik Park
  *Int. Conf. on Computer Vision (**ICCV**), **Main Conference & Demonstrations Track**, 2025* | link

Domestic

- **Language-Driven Robotic Manipulation via Open-Vocabulary 3D Gaussian Scene Understanding**
  Hyunjoon Lee, Eunsung Cha, Jaesik Park
  KCC (Outstanding Paper Presentation Award), 2026
- **Aligning 2D Mask Space Consistency Using 3D Gaussian Splatting**
  Eunsung Cha, **Hyunjoon Lee**, Jaesik Park
  KCC (Outstanding Paper Presentation Award), 2026
- **Efficient Feature Lifting and Compression Using Pre-trained 3D Gaussians**
  Hyunjoon Lee, Joonkyu Min, Jaesik Park
  KCC (Outstanding Paper Presentation Award), 2025

Projects

- **Real2Sim VLA**

  Personal Project

  Fine-tuned a VLA on real-world scene data and reconstructed the same scene with 2DGS into a USD-compatible mesh for Isaac Sim, enabling photorealistic Real2Sim post-training and scene-level evaluation.

- **RAG-based Interactive 3D Car Live Video Service using CF3**

  Kolon Motors, Korea, Sep. 2025 – Dec. 2025

- **End-to-End 3D Reconstruction of Complex Scenes for Robotic Perception**

  Hyundai Motors, Korea, Mar. 2025 – Mar. 2026

- **Robust Pick and Place**

  LG Electronics, Korea, Jun. 2023 – Sep. 2023

- **Radio Station Protection and Frequency Sharing via Spectrum Challenge**

  Institute of Information & Communications Technology Planning & Evaluation, Korea, Jan. 2023 – Jun. 2023

- **Design of a Two-Stage Reducer**

  SKKU, Korea, Mar. 2020 – Jun. 2020

Experiences

- **Academic Collaboration** *May 2026 – Present*

  NVIDIA Taiwan Remote

  ◦ *Collaborator: Jaesung Choe*

  ◦ Beyond the Current Scene: Event-Referential Grasping with Active View Selection

- **Research intern** *Jun. 2023 – Sep. 2023*

  HuGe Lab, KAIST AI Seoul, Korea

  ◦ *Advisor: Beomjoon Kim*

  ◦ Developed a model to predict 6 DoF grasp poses using a depth sensor

- **Research intern** *Jan. 2023 – Jun. 2023*

  CSI Lab, SKKU Suwon, Korea

  ◦ *Advisor: Yusung Kim*

  ◦ Developed an RL agent using the PPO algorithm and transformer architecture

- **Research intern** *Dec. 2019 – Mar. 2020*

  CAMAS Lab, SKKU Suwon, Korea

  ◦ *Advisor: Jachoon Koo*

  ◦ Developed a mathematical model to elucidate the stiffness mechanism of the soft gripper

Awards and honors

- **Outstanding Paper Presentation Award**, "Language-Driven Robotic Manipulation via Open-Vocabulary 3D Gaussian Scene Understanding", Korea Computer Congress, Korea (Aug. 2026)
- **Outstanding Paper Presentation Award**, "Aligning 2D Mask Space Consistency Using 3D Gaussian Splatting", Korea Computer Congress, Korea (Aug. 2026)
- **Outstanding Paper Presentation Award**, Workshop on Image Processing and Image Understanding (**IPIU**), Korea (Feb. 2026)
- **Honorable Mention**, Research Paper Competition, Seoul National University, Korea (Dec. 2025)
- **Outstanding Paper Presentation Award**, "Efficient Feature Lifting and Compression Using Pre-trained 3D Gaussians", Korea Computer Congress, Korea (Aug. 2025)
- **Invited Talks**, "CF3: Compact and Fast 3D Feature Fields" SNU Graduate of AI Research Exchange, Seoul National University, Korea (Jun. 2025)

- **Top Excellence Award at the Creative Research Project**, Seoul National University, Korea (Jun. 2025)
- **Outstanding Award at the demo competition**, Seoul National University, Korea (Jun. 2025)
- **Invited Talks**, "CF3: Compact and Fast 3D Feature Fields", NaverLabs, Korea (Mar. 2025)
- **Dean's List Award**, SungKyunKwan University(SKKU), Korea (May. 2021)
- **Scholarship for academic excellent**, SungKyunKwan University(SKKU), Korea (Mar. 2021)
- **Dean's List Award**, SungKyunKwan University(SKKU), Korea (Oct. 2020)
- **Scholarship for research excellent**, SungKyunKwan University(SKKU), Korea (Mar. 2020)
- **Scholarship for academic excellent**, SungKyunKwan University(SKKU), Korea (Mar. 2019)
- **Scholarship for academic excellent**, SungKyunKwan University(SKKU), Korea (Sep. 2018)

Patent

- Apparatus and Method for Reconstructing Three-Dimensional Shape of Target Object Based on Occlusion Information and Multi-View Images, 10-2026-0192561 (with Hyundai Robotics Lab)
- Method and Apparatus for Optimizing Compact and Fast 3D Feature Fields, 10-2025-0093174

Teacher Assistant

- **Samsung Electronics DS Cooperation Course**, Samsung Electornics (Aug. 2025)
- **Introduction to Artificial Intelligence**, Seoul National University (2025 Spring)
- **Samsung Electronics DS Cooperation Course**, Samsung Electornics (Aug. 2024 – Oct. 2024)
- **Introduction to Artificial Intelligence**, Seoul National University (2024 Spring)